

Xenia

tip the post, not the platform.

On-chain tipping for X (Twitter), built on Somnia Network. Tip anyone by their handle in one click – sub-second finality, sub-cent gas, and an on-chain escrow that lets you tip people before they even have a wallet.

SUBMISSION DOCUMENT • LIVE ON SOMNIA TESTNET

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01 Overview

Xenia turns every post on X (**Twitter**) into a tippable on-chain action. A Chrome extension injects a **Tip** button into the feed and onto profiles; clicking it sends value on **Somnia Network** that either lands instantly in the recipient's wallet or is held in an on-chain **escrow** until the recipient signs up – at which point they claim it with one click.

The entire stack – Solidity contracts, a Node backend, a React dashboard, and a browser extension – is built specifically around Somnia's strengths. Sub-second finality and sub-cent gas make per-post micro-tipping economically viable for the first time, turning a slow, fee-heavy chore into a single in-feed gesture.

In one sentence

Tip anyone on X by their handle, instantly and on-chain – even if they have never touched crypto.

02 The problem

Tipping creators on social media is broken in two directions.

Web2 tipping

- > Both sides must create accounts
- > Platforms take 2-30% in fees
- > Payouts take days
- > Locked to one walled garden

Web3 tipping

- > Recipient needs a wallet on the right chain
- > Gas frequently costs more than the tip
- > No way to tip an off-chain identity
- > Confirmation latency kills the gesture

The net result: tipping is not a fluid, in-feed gesture. It is a friction-filled chore that almost no one actually does.

03 The solution

Xenia removes both walls.

- > **Tip anyone, even non-crypto users.** Tips to an unregistered handle are held in an on-chain escrow keyed by the handle. The recipient signs up once, the backend binds their wallet on-chain, and the funds are theirs.
- > **Tipping is a single click inside X.** The extension injects a Tip button into every tweet and onto profiles, and auto-adds Somnia to the wallet so users never configure a network.
- > **Two modes, one contract.** A direct tip signed by the user's wallet, and a Twitter-command tip executed by an AI bot from a pre-funded, delegated balance.
- > **Handle-first identity.** Every flow – tip, claim, registration – is keyed by the lowercased X handle, the only identifier a sender knows at tip time.

04 How it works

Mode A – direct tip (extension / web)

The user's own wallet signs `Escrow.tip(handle)`. If the recipient is registered, STT transfers directly; otherwise it is escrowed under the handle.

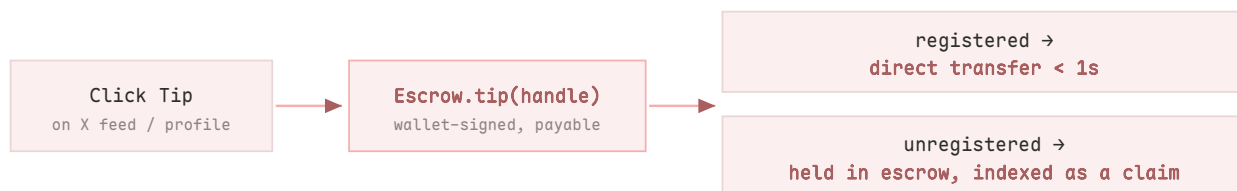


FIGURE 1 – MODE A DIRECT TIP

Mode B – Twitter command

The user pre-deposits and authorizes the bot. They tweet a command; the bot parses intent with Gemini and calls `tipOnBehalf` – it can only forward funds, never withdraw.



FIGURE 2 – MODE B TWITTER COMMAND

Claim

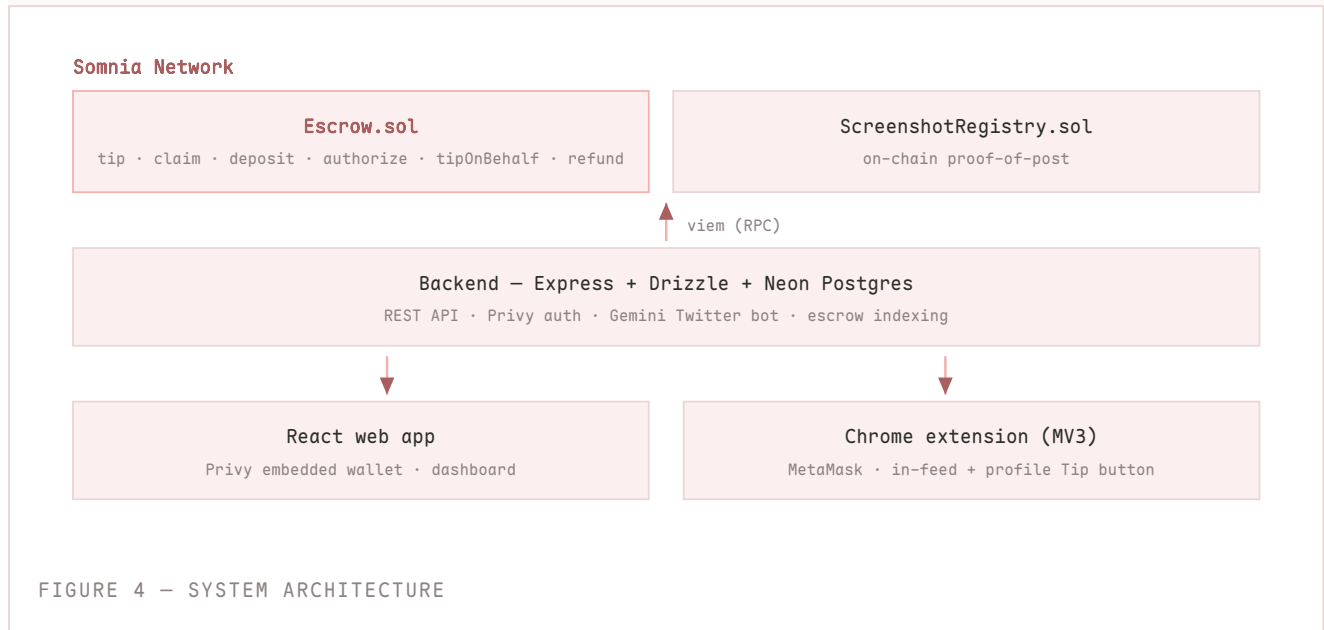
An escrowed recipient signs up, the backend binds their handle to their wallet on-chain (immutable), and they pull all pending tips.



FIGURE 3 – CLAIM FLOW

05 Architecture

One escrow contract on Somnia, reached from four surfaces through a single backend.



The user's wallet signs every value-moving call (tip, claim, deposit, authorize). The backend wallet is the contract owner and only signs registerWallet and registerScreenshot. The bot signs tipOnBehalf from delegated deposits and can never withdraw.

06 Why Somnia

Sub-second finality and sub-cent gas are what make per-post micro-tipping possible at all.

PROPERTY	TYPICAL L1/L2	SOMNIA	WHY IT MATTERS FOR XENIA
Finality	2-12s	< 1s	In-feed tipping feels like Web2
Gas / tip	\$0.05-\$1.00	< \$0.001	\$0.10-\$1.00 tips are economically sane
Throughput	100-4,000 TPS	1,000,000+ TPS	Headroom for a viral app, no fee spikes
EVM	Yes	Yes	Solidity + Hardhat + viem port cleanly

Xenia was originally conceived on a slower chain. The Somnia migration is what enables the consumer-grade UX: sub-second finality changes the gesture from "I'm sending a transaction" into "I just tapped a button."

07 Smart contracts

Escrow.sol

A single contract that handles both tipping modes and the lifecycle of unclaimed funds.

FUNCTION	MODE	DESCRIPTION
<code>tip(handle) payable</code>	A	Direct transfer if registered, else escrow under the handle
<code>deposit() payable</code>	B	Credits the user's internal balance
<code>authorize(delegate)</code>	B	Allow the bot to call <code>tipOnBehalf</code> for this user
<code>tipOnBehalf(sender, handle, amount)</code>	B	Delegate-only; forwards from sender's deposit
<code>claim(handle)</code>	—	Registered wallet pulls all pending escrow tips
<code>refund(handle, idx)</code>	—	Sender refund after a 90-day window
<code>registerWallet(handle, wallet)</code>	—	onlyOwner; immutable once set

ScreenshotRegistry.sol

Lightweight on-chain proof-of-post: `registerScreenshot(cid, tweetId)`, `verifyScreenshot(cid)`, `getProofByTweetId(tweetId)`.

Escrow	0xEf0ca54F3C195737880127df62069C5B5A17B458
Registry	0x9C3c6b9cc4ECdA73e65A240DD0cD075ce202AfE3
Chain	Somnia Shannon Testnet • chainId 50312 • STT

08 Security

- > **Reentrancy guard** on every value-moving external call.
- > **Fee accounting separated** from user funds – fee withdrawals can never touch balances.
- > **Immutable wallet binding** – a handle can be bound to a wallet exactly once.
- > **Scoped bot delegation** – `tipOnBehalf` can only forward to the resolved recipient, never to an arbitrary address, and never withdraw.
- > **90-day sender refund** safety net for tips that are never claimed.
- > **Two-step ownership transfer** on both contracts; platform fee capped at 5%.
- > **No custody of keys** – Privy embedded wallets stay client-side; the backend never holds user private keys.

09 Tech stack & what's built

Stack

- > Solidity + Hardhat (contracts)
- > Node + Express + Drizzle ORM
- > Neon Postgres
- > viem (chain client)
- > React 18 + Vite + Tailwind + Privy
- > Chrome MV3 extension
- > Python bot · Gemini / Claude NLU

Working on testnet

- > Both contracts deployed & live
- > Real on-chain tip / claim / deposit
- > Escrow tips indexed to Postgres
- > Extension: in-feed + profile Tip button
- > Gemini-powered command bot
- > Privy auth (X or wallet)
- > Deploy-ready: Railway + Neon + preflight

10 Roadmap & network reference

Phase	Scope
Now	Live on Somnia testnet · Mode A end-to-end · escrow indexing · extension demo
Next	Mainnet deployment (chainId 50313, SOMI) · Chrome Web Store listing · X login via Privy
Later	@xeniaBot launch on a paid X API tier · indexer-backed leaderboards · contract audit

Network reference

Property	Testnet	Mainnet
Chain ID	50312	50313
Symbol	STT	SOMI
RPC	dream-rpc.somnia.network	mainnet-rpc.somnia.network
Explorer	shannon-explorer.somnia.network	explorer.somnia.network

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